CHERI FAVEL

Chemical Engineer (MEng) | Process Engineering & Industrial R&D

Abu Dhabi, UAE | UAE resident | Available immediately

cherifavel@gmail.com | +971 58 584 5473 | linkedin.com/in/cheri-favel/

researchgate.net/profile/Cheri-Favel | Engineering portfolio



PROFILE

Chemical Engineer (MEng) with experience in process engineering, early-stage process design and industrial R&D across hydrometallurgy, electrochemical recovery and industrial water treatment. Skilled in engineering calculations, process documentation, experimental development and technical analysis, with experience translating results into practical engineering recommendations.

CORE CAPABILITIES

Process engineering: Mass and energy balances, PFDs, P&IDs, equipment sizing support, design bases and equipment specifications.

Process development: Experimental design, electrolysis, electrowinning, selective precipitation, mineral leaching and industrial water treatment.

Engineering analysis and reporting: Process performance evaluation, kinetic modelling, analytical data interpretation and technical recommendations.

PROCESS ENGINEERING & RESEARCH EXPERIENCE

Process & R&D Engineer

Mar 2021 – Jul 2022

Free Radical Process Design (FRPD) | South Africa

- Prepared mass and energy balances, supported process-unit design and equipment sizing, and developed PFDs and P&IDs.
- Prepared process descriptions, equipment specifications and technical reports, contributed to design bases and reviewed supplier quotations.
- Supported laboratory and pilot-scale process development. Contributed to design-basis documentation for a manganese sulphate monohydrate demonstration plant and ran selective-precipitation trials.
- Tested a laboratory-scale gold-recovery reactor on refinery eluate, varying current density, voltage, flow rate and reaction time. Supported mass-balance analysis and interpreted ICP-OES, XRF and pH results.
- Ran copper and cobalt electrowinning trials on a continuous rotating-cathode unit, and contributed to its proposed application for zinc removal from tailings water.
- Planned, set up, ran and reported plate-electrode zinc electrolysis trials end to end, including experimental design, sampling, analysis and recommendations.
- Evaluated oxidation and selective precipitation for Fe, Zn and Mn removal from mine-tailings water, and wrote the laboratory-trial section of a research proposal for a Peruvian tailings-water project.
- Ran lanthanide precipitation experiments and contributed calculations to a CAPEX/OPEX model for a neodymium metal plant.

Postgraduate Researcher in Chemical Engineering

Jan 2020 – Dec 2021

University of Pretoria | South Africa

- Designed nitric-acid batch leaching experiments on phlogopite and characterised the solid residues. Identified a diffusion-controlled mechanism and determined activation energies.
- Analysed kinetic and characterisation data with Python and Excel. Delegated tasks and supervised undergraduate laboratory work. Reported findings in Clays and Clay Minerals (2022), doi.org/10.1007/s42860-022-00180-x.

EDUCATION & SELECTED PROJECTS

Master of Engineering (MEng), Chemical Engineering

University of Pretoria, 2021. Research on phlogopite leaching kinetics and solid-residue characterisation.

BEngHons and BEng, Chemical Engineering

University of Pretoria, 2020 and 2019, respectively. Honours distinction and academic honorary colours.

Academic design projects included a non-adiabatic packed-bed pre-reactor for ethyl acetate reactive distillation and a portable water-purification concept using ultrafiltration and activated carbon.

ADDITIONAL PROFESSIONAL EXPERIENCE

Business Setup & Marketing Consultant Apr 2025 – Present
Cresco Consulting FZCO | Dubai, UAE

- Coordinate branding, website and marketing deliverables and support UAE business-setup documentation.

Engineer Assistant / Deckhand / Stewardess May 2023 – Apr 2025
Research Vessel Rosa | United States & Bahamas

- Operated pumps, valves and onboard water systems as required. Assisted with plumbing and RO watermaker maintenance and operation, and organised technical drawings, maintenance records and procurement.

Academic Journal Editor Nov 2022 – Apr 2023
Cactus Communications | Remote

- Edited chemical engineering and scientific manuscripts for technical clarity, terminology, consistency and journal requirements.

Teaching Assistant Feb 2019 – Jun 2021
University of Pretoria | South Africa

- Supported kinetics and particle technology laboratories, prepared practical sessions, supervised student work and assessed submissions.

Laboratory Intern Dec 2017 – Jan 2019
University of Pretoria | South Africa

- Completed 12 weeks of laboratory work on mineral leaching and fertiliser production, including experimental planning and analytical testing.

TECHNICAL TOOLS

- **Data analysis**: Excel (Power Query, PivotTables and VBA) and Python.
- **Engineering**: Visio and SolidWorks. Foundational exposure to Aspen Plus/HYSYS.
- **Laboratory**: ICP-OES, pH and particle-size analysis. Interpretation of XRF, XRD, ATR-FTIR, BET, TGA and SEM-EDS results.

SELECTED TRAINING & PROFESSIONAL DEVELOPMENT

- Project Management – Energy and Water Projects — PMI & LinkedIn Learning, Feb 2026
- Occupational Safety and Health: Hazard Communication — LinkedIn Learning, Feb 2026
- Approved Engine Course (AEC 1), 2024 · STCW training, 2023
- Operations Management Professional Certificate · Jack Welch Management Institute, LinkedIn Learning · In progress, started September 2026

LANGUAGES

- English (Native) | Afrikaans (Proficient) | Hebrew (Basic)